Semi-mechanistic models for estimating and interpreting long-term changes in infectious disease transmission from historical epidemic time series

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Abstract

Introduction

A fundamental aim in infectious disease dynamics research is characterizing how disease transmission rates change over time [1, 2, 3, 4]. For example, classic studies of endemic childhood infections, particularly measles, focused on estimating seasonal fluctuation in transmission rates driven by school term and climate effects [5, 6, 7, 8]. These efforts not only generated key insights into the drivers of epidemic dynamics but also established core methodological foundations for infectious disease research [9, 1, 10, 11]. Yet transmission rates can also evolve over longer time scales, reflecting gradual shifts in host behavior [12, 13], demography, socioeconomic factors [3], and environmental conditions [14]. Compared with seasonal variation, however, these low-frequency changes have received far less attention, in part because they are more difficult to identify from incidence data alone and are often confounded with other slow-moving processes such as susceptible replenishment and vaccination [2].

Disruptions in disease transmission caused by COVID-19 nonpharmaceutical interventions also highlighted the similar need for new frameworks for estimating transmission rates that can vary flexibly across two different time scales: explaining these disruptions requires simultaneous estimation of seasonal transmission patterns and rapid changes in transmission rate reflecting intervention effects. To capture the impact of nonpharmaceutical interventions on disease transmission, some researchers have relied on known metrics of intervention strengths, such as mobility patterns [15, 16] and stringency indices [17, 18, 19], whereas other have sought to directly estimate these impacts through various statistical methods [20, 21, 22, 23]. Notably, models utilizing Bayesian inference framework have proven particularly useful for estimating flexible changes in transmission patterns, where statistical constraints through prior distributions can appropriately limit the variability in parameter estimates [20, 24, 25, 26]. These methodological advances provide a basis for developing semi-mechanistic models that allow estimation of transmission rates across two different time scales, with potential applications beyond the context of COVID-19 interventions, but their estimability and generalizability have not yet been tested. Furthermore, the apparent discrepancy between the estimated intervention effects and observed changes in mobility patterns indicate potential challenges in interpreting parameter estimates from semi-mechanistic models [24].

Inspired by these efforts, we present a Bayesian semi-mechanistic framework for simultaneously estimating seasonal and long-term variation in disease transmission and investigate its estimability and interpretibility. We test the resulting framework against a 126-year long time series of smallpox deaths, which presents a computationally challenging example. We then validate the methodology using simulation-recovery experiments.

Results

Developing semi-mechanistic models for estimating seasonal and long-term variation in disease transmission

First, the data set consisted of weekly reports of smallpox deaths spanning 126 years (Figure 1A). During this period, many factors that could affect disease transmission changed, including variolation practices and demographic patterns. For instance, variolation was first introduced in 1721 and increased in intensity until the late 1790s and early 1800s, when vaccination was introduced. These changes likely decreased smallpox transmission, but quantifying the prevalence and effectiveness of variolation practices is challenging. During the same period, the birth rate also increased, contributing to an almost 2.5-fold increase in population size, which likely increased contact rates and smallpox transmission. Many other changes likely occurred, including changes in sanitary practices and population structure.

It would be infeasible to explicitly incorporate all these changes into a model, especially because quantitative data describing the underlying changes are unavailable. Instead, our goal was to develop a semi-mechanistic model that seeks to implicitly capture the effects of unmodeled factors on disease transmission (Figure 1B). Specifically, we decomposed the transmission rate $\beta(t)$ of the classic SIR model into a seasonal term $\beta_{\text{seas}}(t)$ and a slow, time-varying term $\delta(t)$:

$$\beta(t) = \frac{\beta_{\text{seas}}(t)\delta(t)}{N_{1670}}, \tag{1}$$

where N_{1670} represents the population size at the beginning of 1670, which is included to scale the $\beta_{\text{seas}}(t)$. Note that we are assuming density dependence transmission to account for potential impact of increase in population sizes on transmission. For the main analysis, we allowed $\delta(t)$ to vary every year and considered slower variation as a supplementary analysis **[SWP:** *TODO*]. Both $\beta_{\text{seas}}(t)$ and $\delta(t)$ terms were modeled using random walk priors to impose smoothness. We then try to fit the resulting SIR model to the time series of smallpox deaths and try to estimate both terms simultaneously using Bayesian inference.

Fitting the resulting model to 126 years of weekly observations presents substantial computational challenges. To address these challenges, we obtained maximum a posteriori estimates through optimization and used a Laplace approximation to characterize posterior uncertainty. Implementation details are provided in the Materials and Methods section.

Potential long-term changes in smallpox transmission

Our semi-mechanistic SIR model was able to capture the overall epidemic patterns across the 126-year time series, including long-term changes in outbreak magnitude (Figure 2A). In particular, we estimated a clear seasonal pattern in transmission,

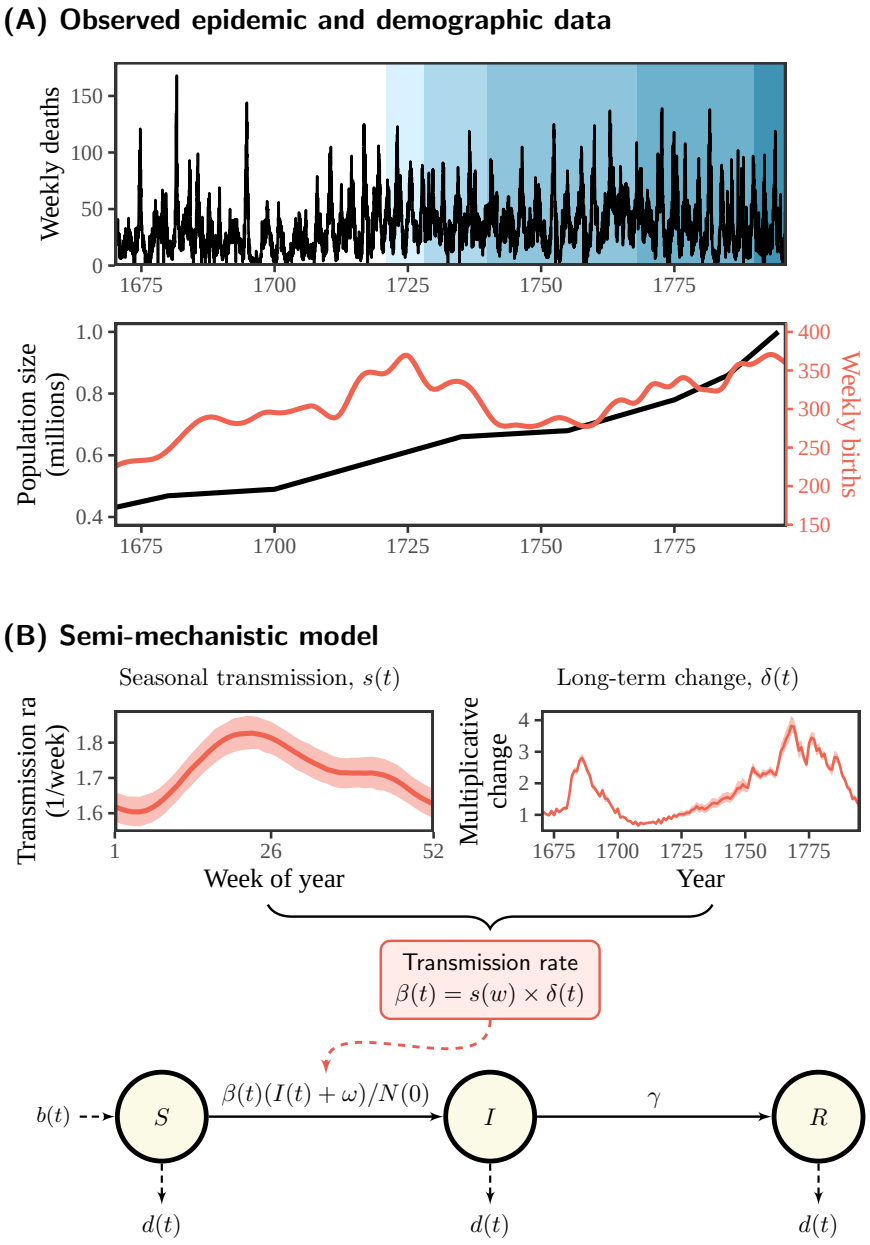


Figure 1: **Developing semi-mechanistic models for estimating long-term changes in transmission from historical epidemic time series data.** (A) Observed epidemic and demographic data, including population size and birth rate. Blue shaded regions indicate the increasing intensity of variolation practices. (B) Schematic diagram illustrating the semi-mechanistic model. The semi-mechanistic model allows for slow, annual variation and fast, seasonal variation in disease transmission rates.

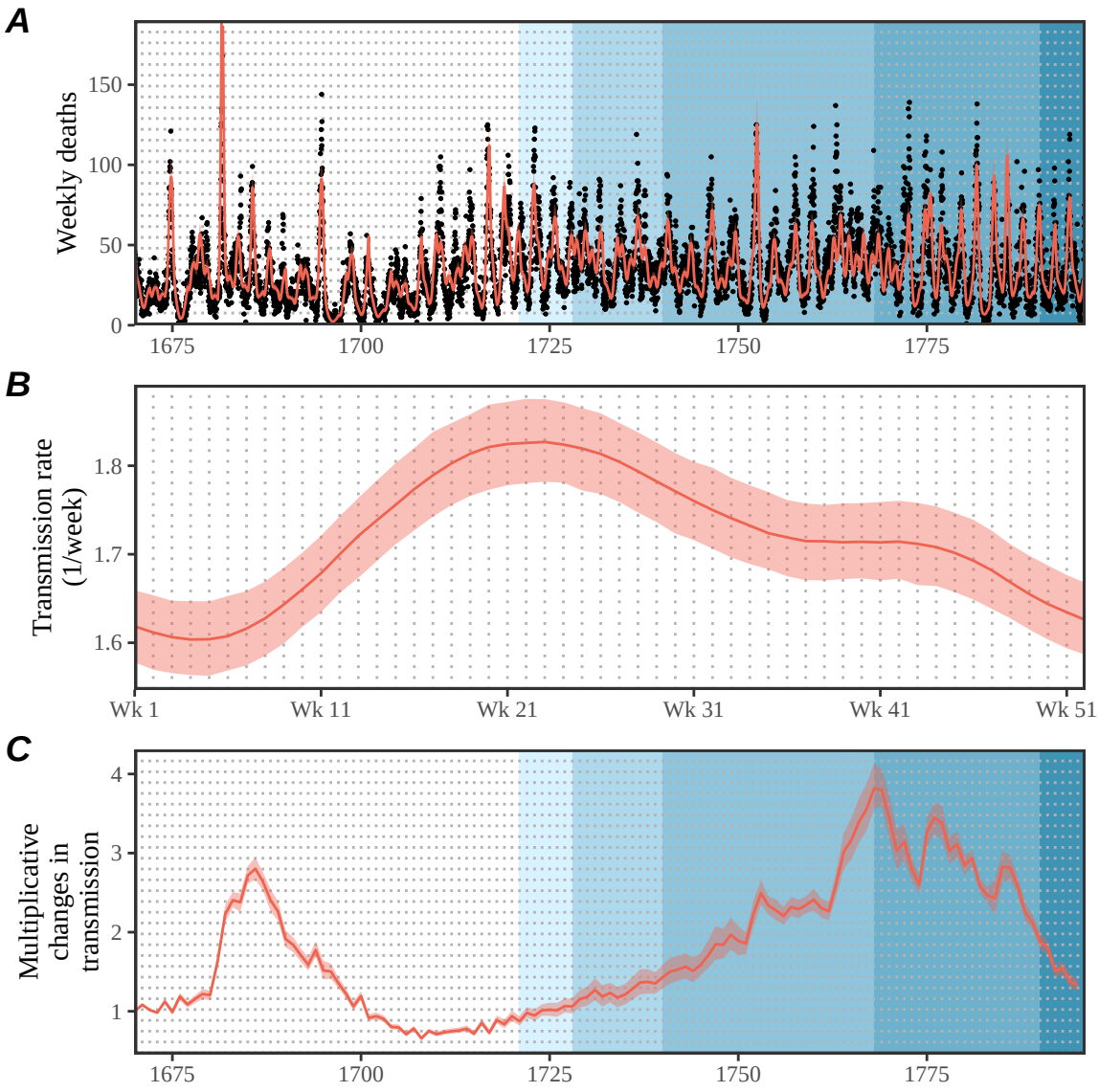


Figure 2: **Summary of model fits to smallpox data set.** (A) Observed data (black points) and model fits (red lines). (B) Estimated seasonal transmission rate. (C) Estimated multiplicative changes in annual transmission. Red lines indicate the posterior median estimate and shaded regions represent the 90% credible interval. Blue shaded regions indicate the increasing intensity of variolation practices.

with transmission increasing from the beginning of the year, peaking around week 23, and then gradually decreasing toward the end of the year (Figure 2B).

We also estimated substantial long-term variation in transmission rates (Figure 2C). The annual transmission term increased sharply during the 1680s, reaching nearly three times its value in 1670, and then declined to its lowest value around early 1700s. Even after the introduction of variolation in 1721, the annual transmission

multiplier gradually increased again, reaching almost four times its 1670 value in the late 1760s. Despite the continued increase in the intensity of variolation practices, annual transmission remained relatively high throughout the 1770s and 1780s before decreasing rapidly toward the end of the study period. This apparent discrepancy suggests that the estimated annual transmission term likely captures the combined effects of multiple unmodeled factors rather than the effect of variolation alone.

Simulation-recovery experiments

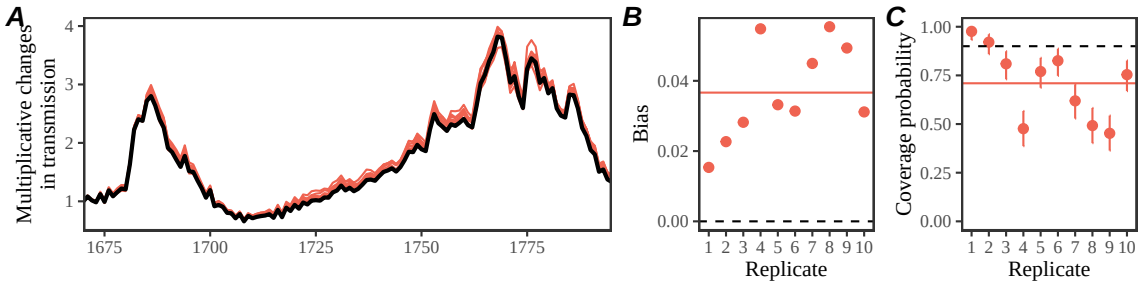


Figure 3: **Summary of simulation-recovery experiments.** (A) Comparisons between the assumed (black) and estimated (red) changes in annual transmission rates. (B) Mean bias across estimated changes in annual transmission rates for each replicates. The black dashed line represents zero bias. The red line represents the average value across replicates. (C) Mean coverage probability across estimated changes in annual transmission rates for each replicates. The black dashed line represents the nominal coverage probability (0.9). The red line represents the average value across replicates.

The discrepancy between increasing variolation efforts, which we expected to drive reduction in transmission, and the estimated patterns of increasing transmission further raises a question of whether the semi-mechanistic model is able to correctly estimate the underlying transmission pattern. Thus, we performed simulation recovery experiments, where we generated 10 synthetic time series using the estimated values and fitted the same model to each time series independently (Figure 3). We found that our model is able to accurately recover long-term changes in transmission (Figure 3A). Across all 10 simulations, the model was able to accurately capture the observed epidemic patterns with R^2 values over 0.7 consistently. However, our methodology introduced some bias ($\approx$ 4%; Figure 3B) and provided low coverage values ($\approx$ 0.7; Figure 3C), suggesting overly narrow credible intervals.

As a negative control experiment, we also generated 10 synthetic time series, where the annual term $\delta(t)$ is set to 1, and fit the same model to each time series independently (Figure 4). Since we assumed random walk priors, the estimated long-term changes in transmission diverged from one, leading up to 20% estimated changes in transmission rates over 126 years (Figure 4A). Despite this divergence, our

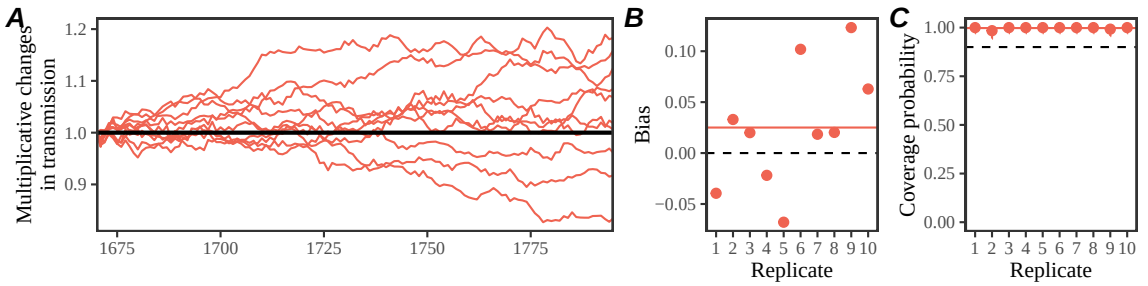


Figure 4: **Summary of simulation-recovery experiments in the absence of long-term changes in transmission.** (A) Comparisons between the assumed (black) and estimated (red) changes in annual transmission rates. (B) Mean bias across estimated changes in annual transmission rates for each replicates. The black dashed line represents zero bias. The red line represents the average value across replicates. (C) Mean coverage probability across estimated changes in annual transmission rates for each replicates. The black dashed line represents the nominal coverage probability (0.9). The red line represents the average value across replicates.

estimates still did not generate any meaningful patterns and provided high coverage values (≈ 1), indicating that credible intervals consistently includes one and that deviation from one is not statistically meaningful (Figure 4C). For this experiment, the model also gave similar but slightly smaller levels of bias ($\approx 2\%$; Figure 4B).

Discussion

Here, we developed a Bayesian semi-mechanistic approach for estimating long-term changes in transmission from incidence time series alone. We demonstrated its application using historical time series on smallpox deaths and simulations, which span over 120 years. In particular, simulation-recovery experiments demonstrated the feasibility of accurately estimating long-term changes in transmission.

Materials and Methods

Data

We used a weekly time series of smallpox deaths to test the ability of a semi-mechanistic model to infer long-term changes in disease transmission (Figure 1). The original data set spans from 1664 to 1930 and was digitized from London Bills of Mortality and the Registrar General's Weekly Returns by [27]. Here, we considered mortality records between 1670 and 1795 to focus on the dynamics before the discovery of vaccination. Even in the absence of vaccination, an increase in variolation to protect against severe outcomes (Figure 1A) and a rapid increase in population

sizes due to increasing births (Figure 1B,C) would have caused complex changes in transmission conditions over the 126-year period, which cannot be explained by traditional modeling approaches that only account for seasonal transmission forcing. Data on population sizes and birth rates were obtained from the publicly available data set from [27]. Death rates were calculated to match increasing rates of population sizes: specifically, we calculated the difference between the weekly rate of population size increase and weekly birth rate to obtain weekly death rates.

Mathematical model

To focus on the feasibility of estimating long-term changes in transmission, we decided to use the simplest transmission model, namely the SIR model. Assuming density dependent transmission, the discrete time SIR model can be written as:

$$\begin{aligned}
\mathrm{FOI}(t) &= \beta(t)(I(t)+\omega) && (2)\\
\Delta S(t) &= \left[1-\exp(-(\mathrm{FOI}(t)+d(t))\Delta t)\right] S(t-\Delta t) && (3)\\
N_{SI}(t) &= \frac{\mathrm{FOI}(t)\Delta S(t)}{\mathrm{FOI}(t)+d(t)} && (4)\\
\Delta I(t) &= \left[1-\exp(-(\gamma+d(t))\Delta t)\right] I(t-\Delta t) && (5)\\
N_{IR}(t) &= \frac{\gamma \Delta I(t)}{\gamma+d(t)} && (6)\\
S(t) &= S(t-\Delta t)+b(t)N-\Delta S(t) && (7)\\
I(t) &= I(t-\Delta t)-\Delta I(t)+N_{SI}(t) && (8)
\end{aligned}$$

where $\mathrm{FOI}(t)$ represents the force of infection; $\beta(t)$ represents the transmission rate; ω represents the number of imported infections; γ represents the recovery rate; $d(t)$ represents the death rate; and $b(t)$ represents the birth rate. The transmission rate is decomposed into a seasonal term $\beta_{\text{seas}}(t)$ and a multiplicative time-varying transmission term $\delta(t)$:

$$\beta(t) = \frac{\beta_{\text{seas}}(t)\delta(t)}{N_{1670}}, \tag{9}$$

where N_{1670} represents the population size at the beginning of 1670, which is included to scale the $\beta_{\text{seas}}(t)$. Previous studies have illustrated the feasibility of estimating $\delta(t)$ at different time scales by anchoring the seasonal term $\beta_{\text{seas}}(t)$ to certain periods. For example, when inferring the impact of COVID-19 interventions, $\delta(t)$ was set to 1 during the pre-pandemic period and allowed to vary from 2020 onward, with change points defined at intervals ranging from one to four weeks [24, 25, 26]. Since we are interested in capturing long-term changes in transmission, we allowed $\delta(t)$ to vary every year.

Statistical inference and implementation

In order to simultaneously estimate $\beta_{\text{seas}}(t)$ and $\delta(t)$ alongside other parameters and initial conditions, we needed to appropriately constrain the parameter space. Otherwise, estimating 52 terms for $\beta_{\text{seas}}(t)$ and 126 terms for $\delta(t)$ would require estimation of > 170 parameters, which is more than an order of magnitude greater than the number of parameters that are estimated for a typical infectious disease model. Bayesian inference, implemented via the Stan software, enables us to constrain the parameter space through prior distributions and allows for fast, efficient inference through state-of-art inference algorithms [28]. We first discuss our model parameterization, prior choices, and observation models used for fitting. Then, we describe our implementation in Stan.

If we were to estimate all 52 terms of the seasonal term $\beta_{\text{seas}}(t)$ separately, then the resulting transmission rate may not be realistic and difficult to interpret. Thus, to constrain the shape of the seasonal term $\beta_{\text{seas}}(t)$ and allow for biological interpretation, we imposed a cyclic, random-walk prior on a log scale [26]:

$$\log \beta_{\text{seas}}(t+1) \sim \text{Normal}(\log \beta_{\text{seas}}(t), \sigma_{\text{seas}}), \quad t = 1, \ldots, 51 \tag{10}$$

$$\log \beta_{\text{seas}}(1) \sim \text{Normal}(\log \beta_{\text{seas}}(52), \sigma_{\text{seas}}) \tag{11}$$

where the standard deviation σ_{seas} determines the smoothness of the seasonal transmission rate. Strong priors were imposed to constrain the parameter space for σ_{seas}:

$$\sigma_{\text{seas}} \sim \text{Half-Normal}(0, 0.01). \tag{12}$$

We further imposed an informative prior on $\beta_{\text{seas}}(t)$, reflecting previously reported $\mathcal{R}_0$ values for smallpox transmission:

$$\log \beta_{\text{seas}}(t) \sim \text{Normal}(0.5, 0.1). \tag{13}$$

Combined with the assumed recovery rate of $-\log(1-1/3)$/weeks, which corresponds to a mean infectious period of 3 weeks, the prior distribution of the basic reproduction number has a median of 4.9 and a 95% quantile of 4.1 to 6.0.

For similar reasons as above, we also utilized random walk priors to model the multiplicative time-varying transmission term on a log scale: $\delta(t) = \exp(\epsilon(t))$. In order to reduce correlations in posterior distributions, we parameterized $\epsilon(t)$ by expression it as a sum of independent normal variables $z_\epsilon(t)$:

$$z_\epsilon(t) \sim \text{Normal}(0, 1), \quad t = 1671, \ldots, 1795 \tag{14}$$

$$\sigma_\epsilon \sim \text{Half-Normal}(0, 0.01) \tag{15}$$

$$\epsilon(1670) = 0 \tag{16}$$

$$\epsilon(t) = \epsilon(t-1) + \sigma_\epsilon z_\epsilon, \quad t = 1671, \ldots, 1795 \tag{17}$$

$$\delta(t) = \exp(\epsilon(t)) \tag{18}$$

where $z_\epsilon(t)$ determines the smoothness of $\epsilon(t)$. By setting the initial $\epsilon(1670)$ to zero and imposing constraining priors on σ_ϵ, this approach anchors the seasonal

transmission term $\beta_{\text{seas}}(t)$ to early periods and allow for slow variation in transmission rates over time.

The model was fitted to smallpox deaths $D(t)$ by assuming a negative binomial observation model:

$$D(t) \sim \text{NegBin}(\rho N_{IR}(t), \phi) \tag{19}$$
$$\rho \sim \text{Beta}(1, 1) \tag{20}$$
$$\phi \sim \text{Normal}(100, 20) \tag{21}$$

where ρ represents the infection mortality rate, and ϕ represents the overdispersion parameter. Alongside parameters described earlier, we also estimated the initial conditions $S(1670)$ and $I(1670)$ at the beginning of 1670 as well as imported infections ω:

$$S(1670) \sim \text{Beta}(1, 1) \tag{22}$$
$$I(1670) \sim \text{Half-Normal}(0, 0.001) \tag{23}$$
$$\omega \sim \text{Half-Normal}(0, 200) \tag{24}$$

Previous studies primarily relied on Hamiltonian Monte Carlo (HMC) sampling implemented in Stan to estimate parameters of transmission models [24, 25, 26]. While this approach enables efficient exploration of high-dimensional parameter spaces and provides accurate uncertainty quantification, inference can remain computationally demanding for nonlinear models with many parameters. During our investigation, we also encountered convergence issues, partly reflecting the sensitivity of HMC sampling to initial conditions. Instead, we began with an optimization from 100 random starting values to obtain the maximum a posteriori estimates. Then, we used Laplace approximation to obtain approximate posterior distributions.

Simulation studies

To evaluate the ability of a semi-mechanistic model to correctly infer seasonal and long-term variation in transmission rates, we generated a synthetic data set using median posterior estimates from the smallpox analysis. We then fitted the same model using the same process as outlined above, using 100 random starting values for optimization. We repeated the analysis 10 times.

As a negative control, we also simulated the model without long-term changes in transmission rates, thus fixing $\epsilon = 0$. We then fitted the same model using the same process as outlined above, using 100 random starting values for optimization, repeating the analysis 10 times. For both simulation studies, we compute relative bias and coverage probabilities (using 90% credible interval) for the estimated long-term changes in transmission.

Supplementary Figure

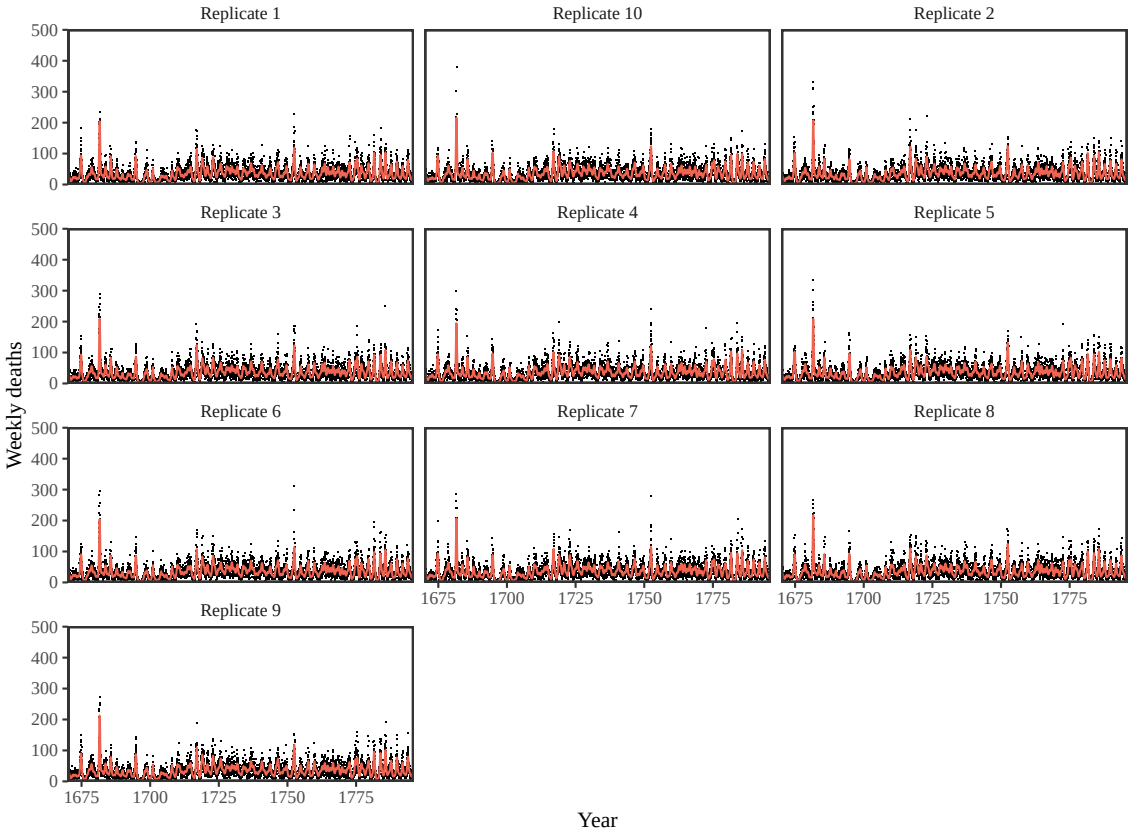


Figure S1: **Comparisons between observed and predicted epidemic trajectories from simulation recovery experiments.**

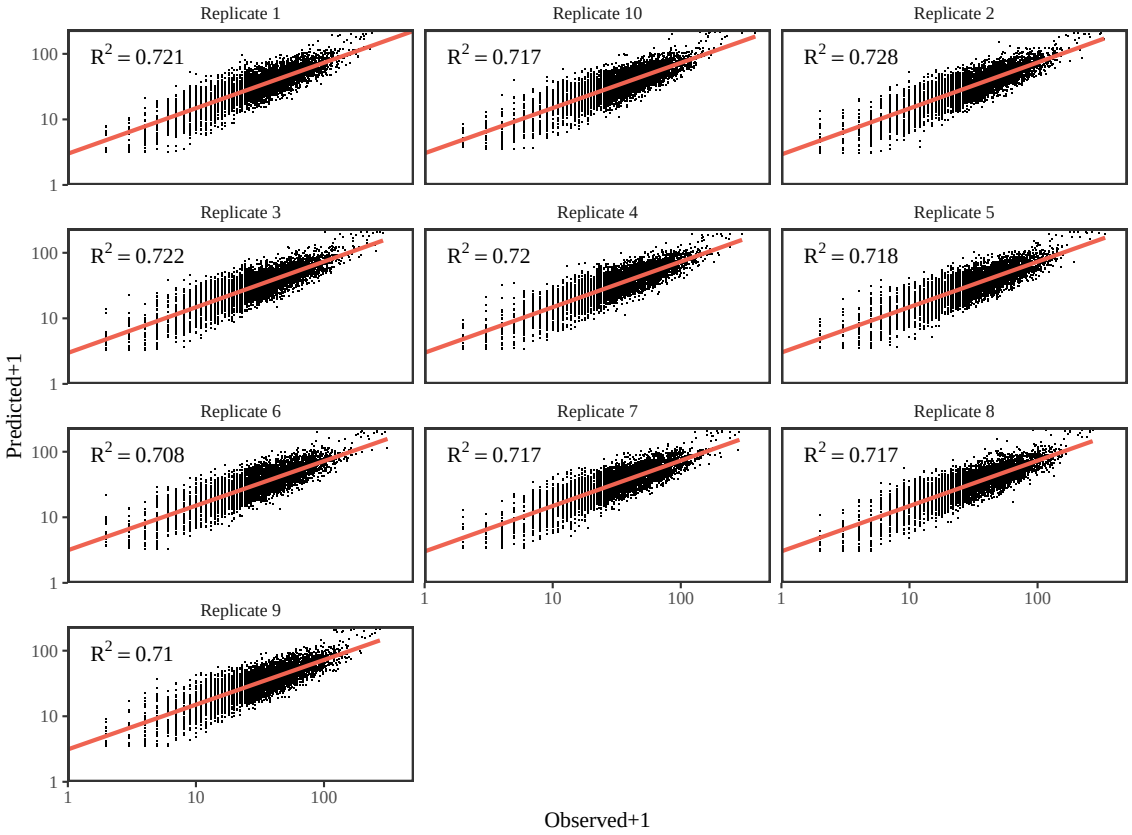


Figure S2: **Summary of comparisons between observed and predicted epidemic trajectories from simulation recovery experiments.**

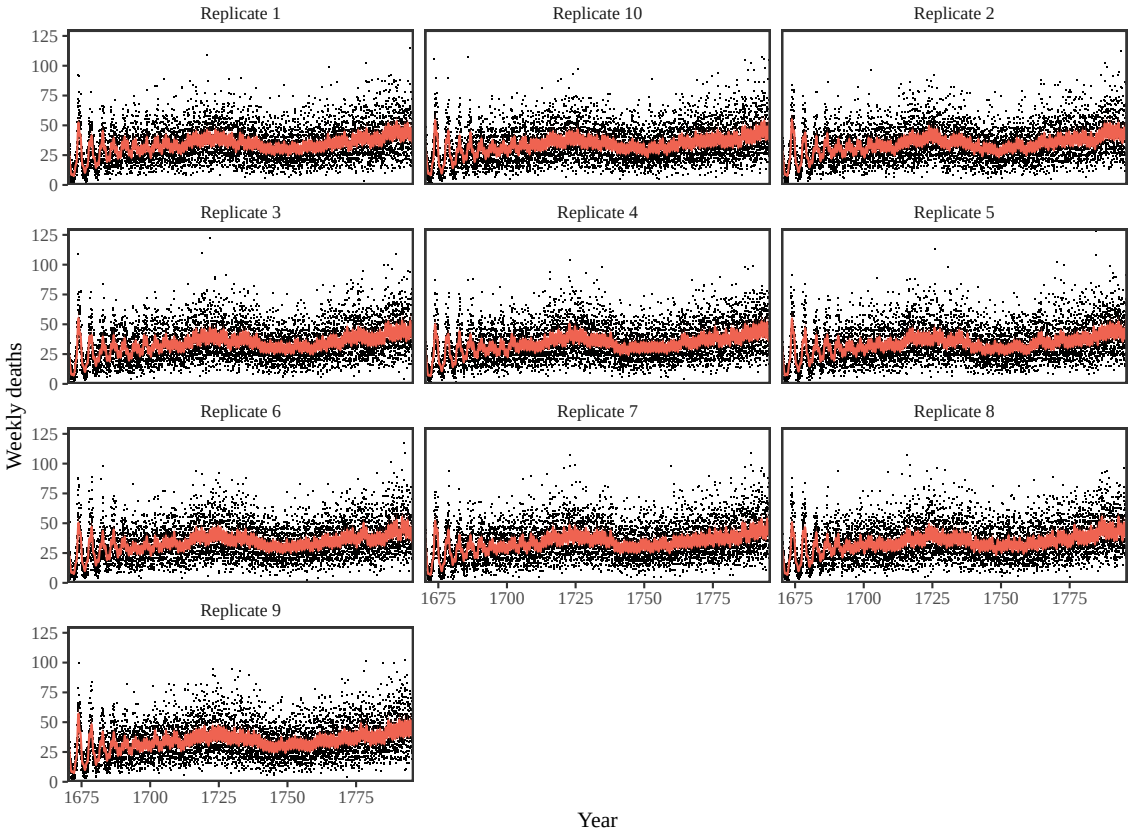


Figure S3: **Comparisons between observed and predicted epidemic trajectories from simulation recovery experiments in the absence of long-term changes in transmission.**

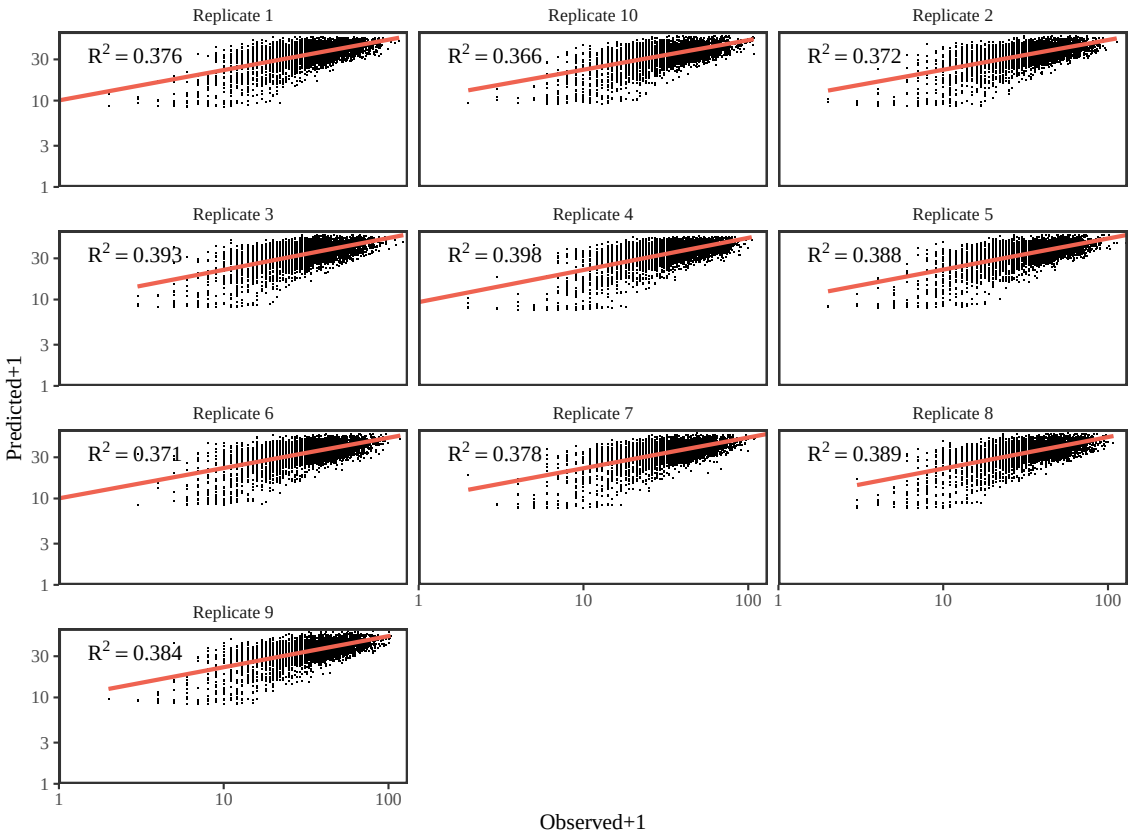


Figure S4: **Summary of comparisons between observed and predicted epidemic trajectories from simulation recovery experiments in the absence of long-term changes in transmission.**

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